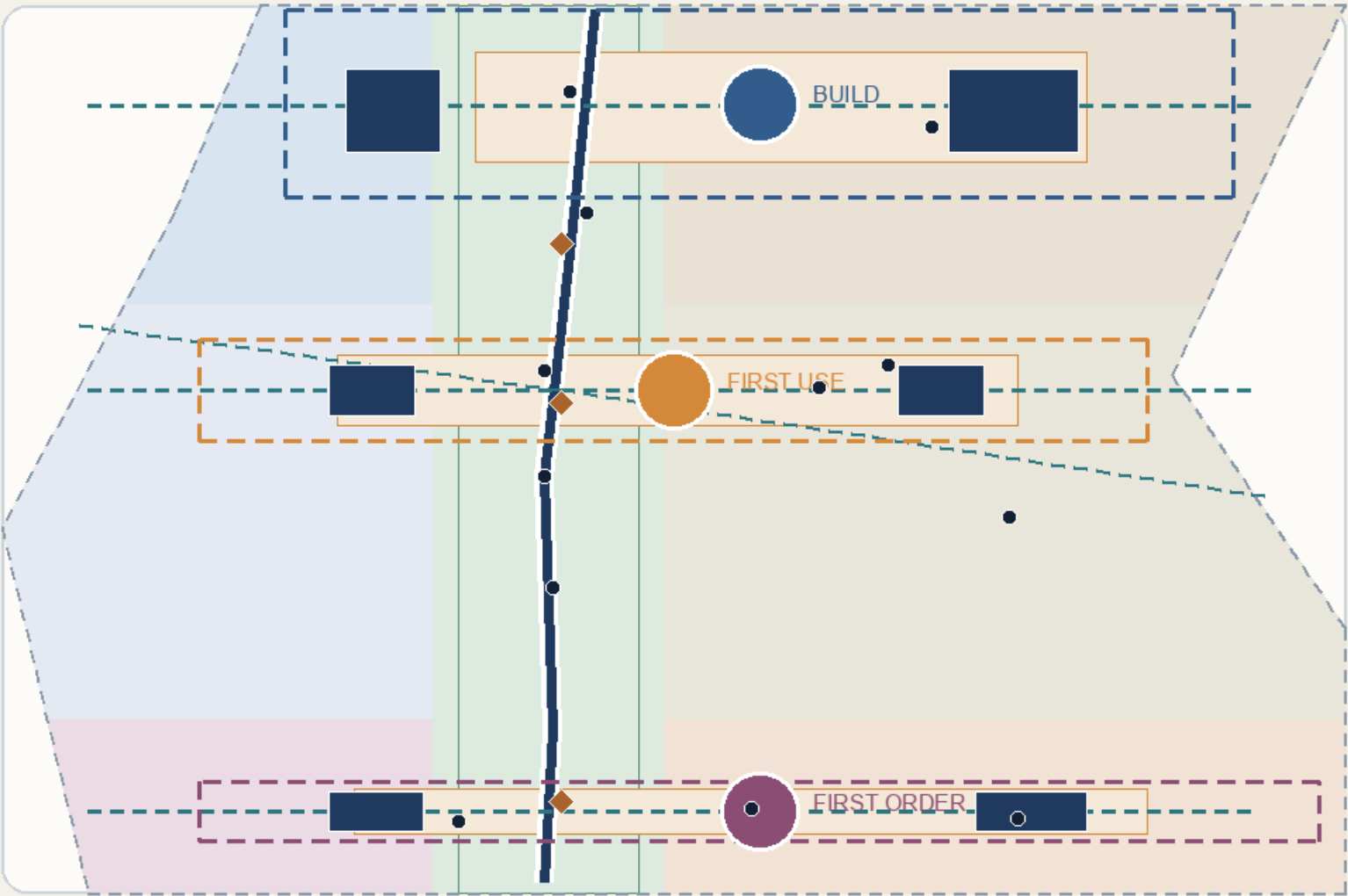


JING-ZHANG FIRST MILE

A governed route from the laboratory to the first real user, first operable scenario, first verifiable order and first verified reuse

JING-ZHANG FIRST MILE: FROM LAB TO REAL ADOPTION

ONE SPINE · THREE STATIONS · TWO WINGS · FOUR OUTCOMES · FIVE GATES



FOUR FIRST-MILE OUTCOMES

- 01 First real user
- 02 First operable scenario
- 03 First verifiable order
- 04 First verified reuse

FIVE LAUNCH GATES

- G0 Real need
- G1 Product ready
- G2 First use
- G3 First adoption
- G4 First reuse

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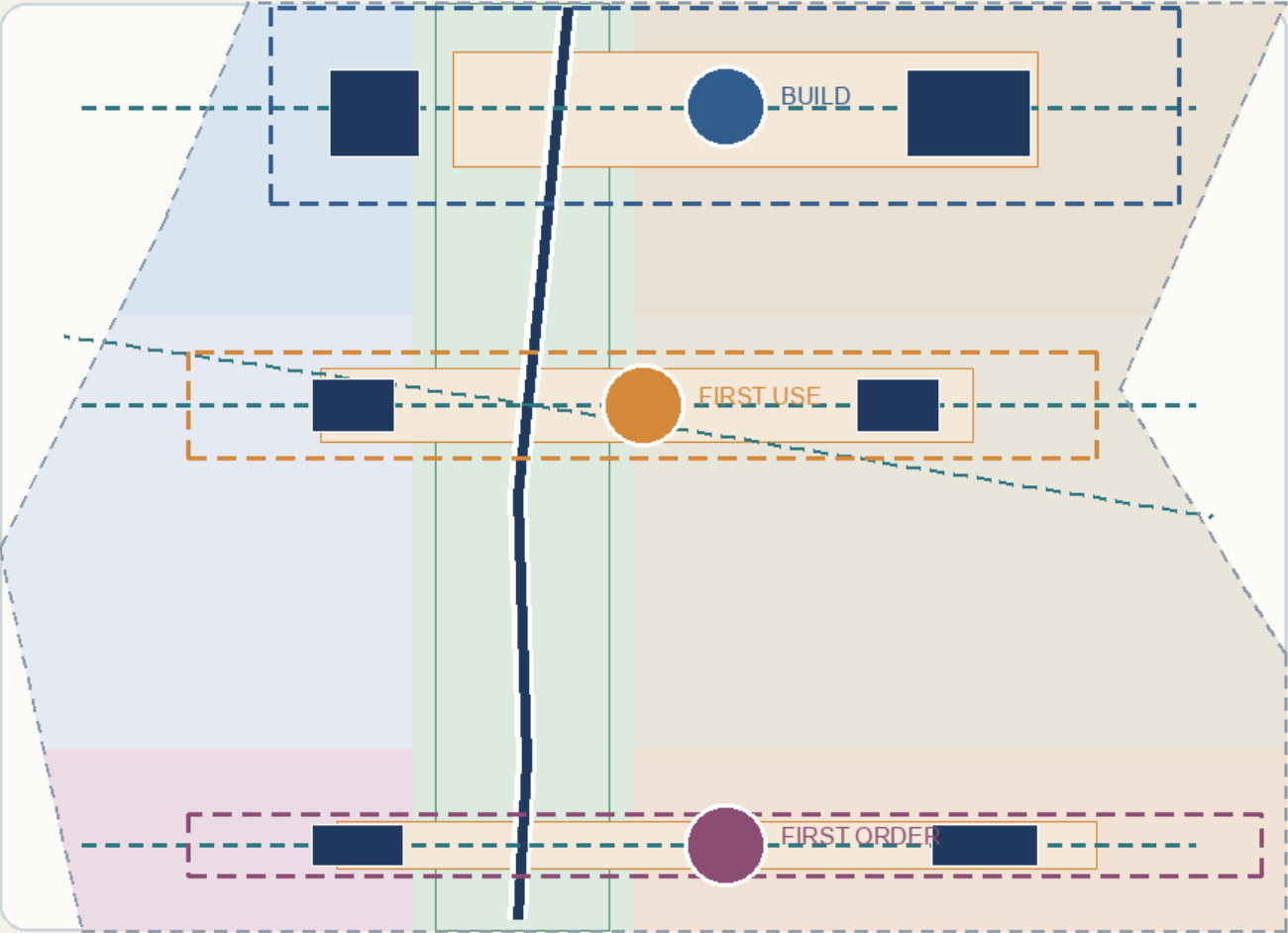
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One Line, Three Stations, Two Wings - One Continuous Handoff

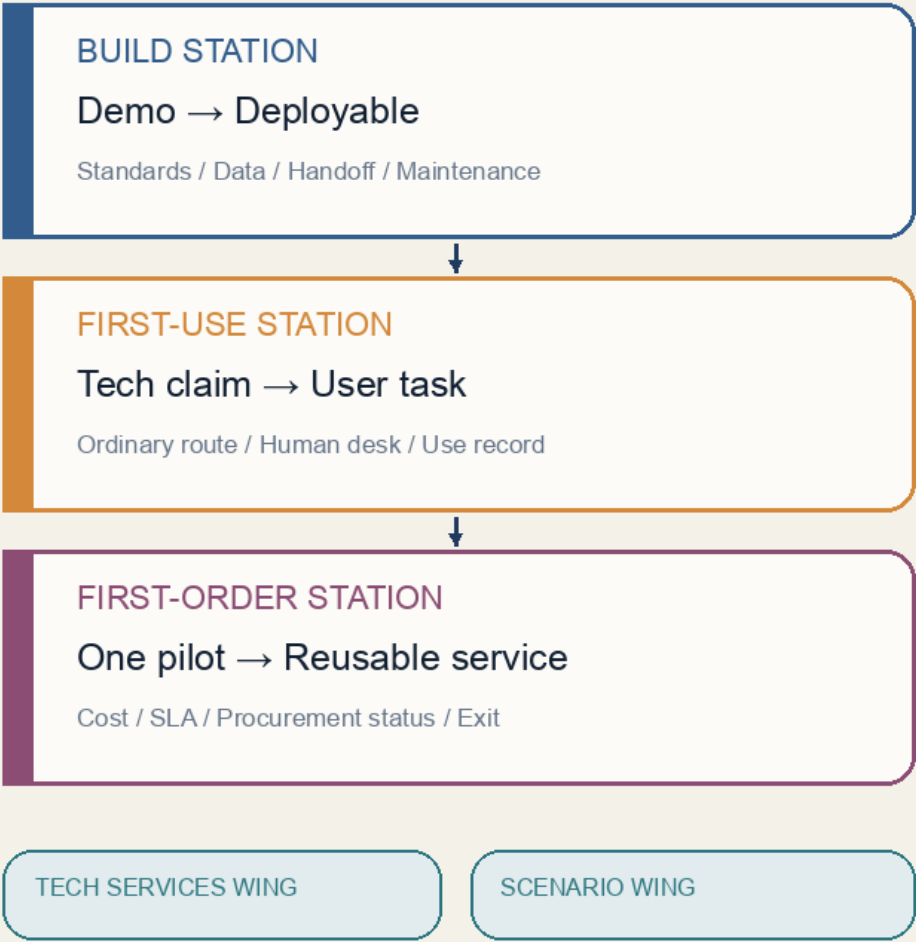
Build proves readiness, First Use proves adoption, and First Order proves operations and reuse; the two wings supply professional services and real scenarios.

SPATIAL STRUCTURE: THREE STATIONS AS A HANDOFF CHAIN

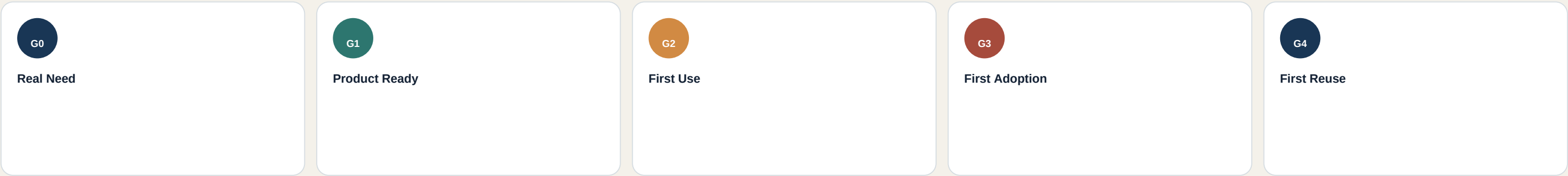
LAND USE AS BACKGROUND; THE DESIGN FOCUSES ON BUILD—USE—OPERATE HANDOFFS



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From Test Capacity to Adoption Capacity

Seven global cases support method, not statutory spatial control.

01 / Punggol

Digital-twin testing before live-district deployment

02 / Helsinki

Companies, researchers, city teams and residents co-test for reuse

03 / Tallinn

One city entry point links departments, sites, data and services

04 / Boston

A small internal city team is the front door for community problems

05 / Woven City

Residents and inventors iterate through ideation, testing and improvement

06 / European Union

Experimentation spaces, living labs and innovation procurement work together

07 / Beijing

Demand, capability and demonstration lists connect first use to orders

08 / Synthesis

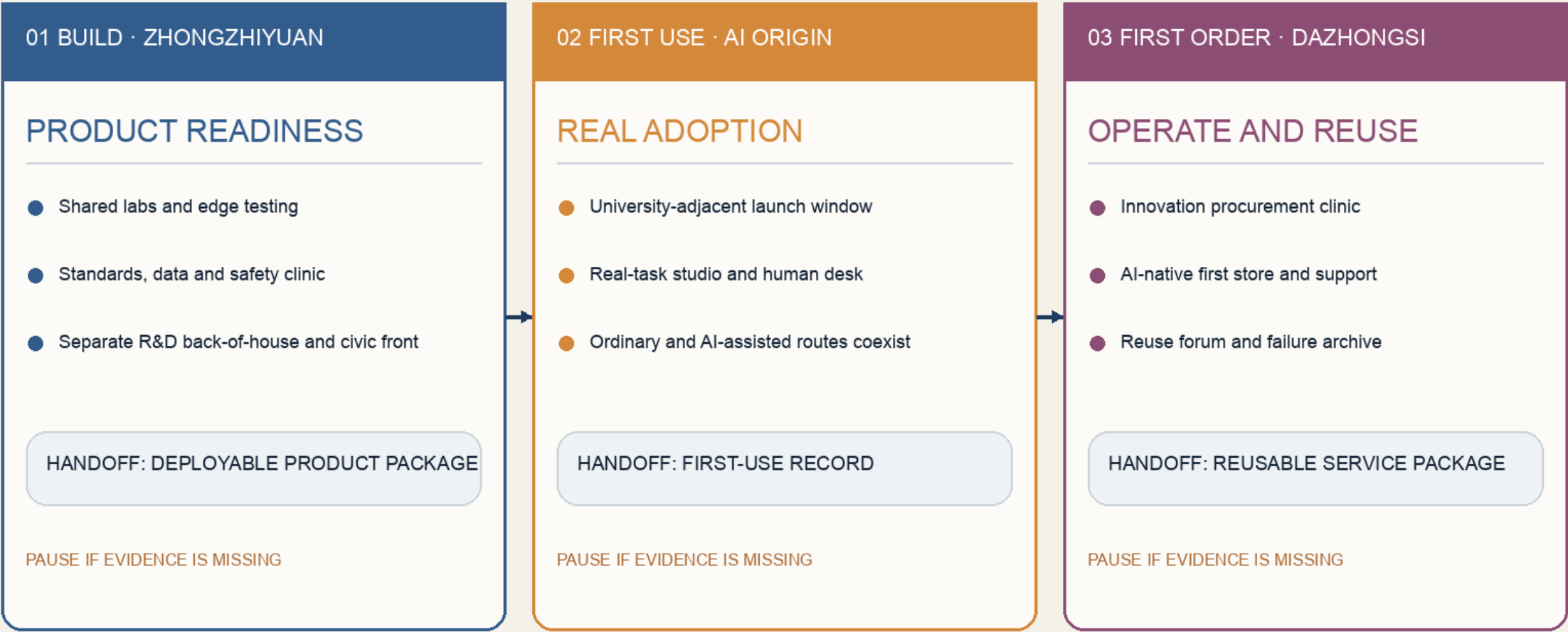
The urban-test bottleneck is not equipment; it is ownership, adoption, handoff and reuse evidence

Three Key Areas, Three Required Delivery Gates

Key-area extents are provisional; all spatial moves are concept recommendations for professional development.

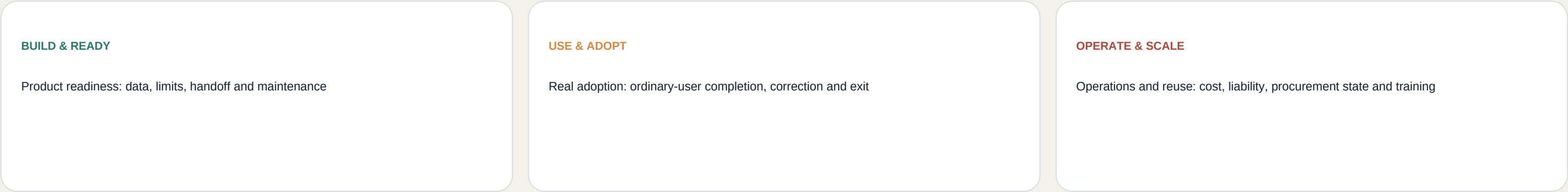
THREE KEY AREAS: EACH STATION MUST HAND OFF TO THE NEXT

ALL EXTENTS ARE PROVISIONAL; SPATIAL MOVES ARE CONCEPTUAL RECOMMENDATIONS



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Twelve Scenario Cards: A Scenario Is Not a Slogan

Every card needs a real problem owner, non-AI baseline, minimum data, human handoff, cost status and stop condition.

01

Urban Problem Launch Desk

Real-need intake · Human moderation

REAL TASK · HUMAN HANDOFF · PAUSE · EXIT

02

Pre-Street Model Evaluation

Fact, bias and safety test · Expert final review

REAL TASK · HUMAN HANDOFF · PAUSE · EXIT

03

Agent Operations Handoff Pod

Unfamiliar operator takeover · Return on failure

REAL TASK · HUMAN HANDOFF · PAUSE · EXIT

04

AI Startup Service Copilot

Public-information navigation · Human correction

REAL TASK · HUMAN HANDOFF · PAUSE · EXIT

05

Accessible Slow-Mobility Task Chain

Barrier work-order closure · Paper and phone remain

REAL TASK · HUMAN HANDOFF · PAUSE · EXIT

06

Multilingual Railway Culture Guide

Human historical review · No route tracking

REAL TASK · HUMAN HANDOFF · PAUSE · EXIT

Twelve Scenario Cards: A Scenario Is Not a Slogan

Every card needs a real problem owner, non-AI baseline, minimum data, human handoff, cost status and stop condition.

07

Public-Space Microclimate Assistant

Environmental advice · Site staff decide

REAL TASK · HUMAN HANDOFF · PAUSE · EXIT

08

Low-Speed Robot First-Use Route

Bounded place, time and speed · Emergency stop

REAL TASK · HUMAN HANDOFF · PAUSE · EXIT

09

AI-Native First Store

Product, after-sales and reuse · No profile pricing

REAL TASK · HUMAN HANDOFF · PAUSE · EXIT

10

Innovation Procurement Clinic

Acceptance, cost and liability · No award promise

REAL TASK · HUMAN HANDOFF · PAUSE · EXIT

11

First Mile Console

Gate, owner and pause state · Public view de-identified

REAL TASK · HUMAN HANDOFF · PAUSE · EXIT

12

Reuse Stage and Failure Archive

Reuse conditions and failure lessons · Anonymous allowed

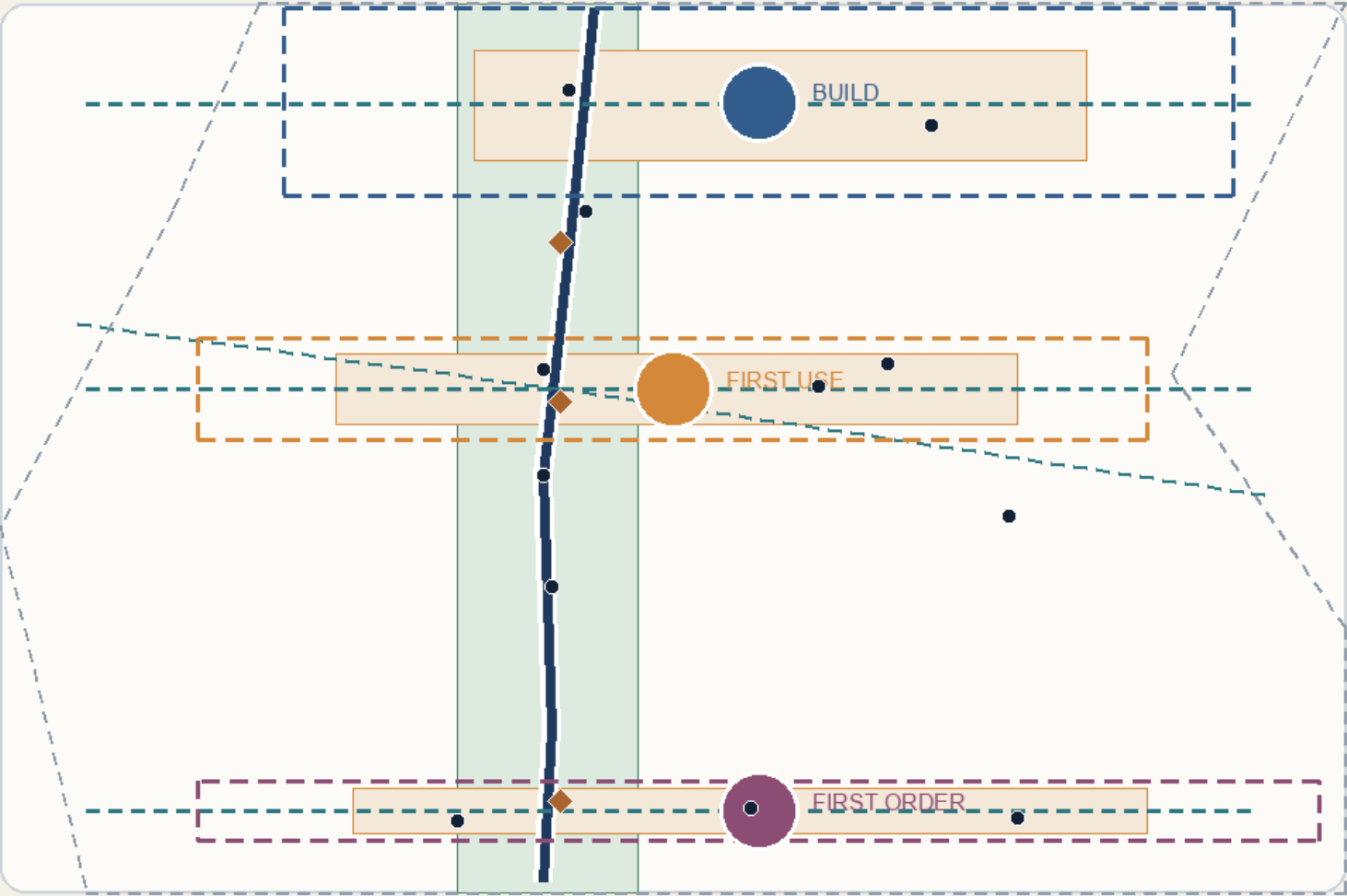
REAL TASK · HUMAN HANDOFF · PAUSE · EXIT

Public Space Still Works When AI Is Removed

Ordinary paths, accessibility, shade, rest, staffed service and non-commercial use always remain primary.

MOBILITY + BLUE-GREEN: PUBLIC VALUE SURVIVES AI REMOVAL

THE ORDINARY ROUTE IS THE BASE; AI REMAINS OPTIONAL, OVERRIDABLE AND REVERSIBLE



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FIVE-LAYER PUBLIC SECTION

- 1 BLUE-GREEN + SHADE
- 2 CONTINUOUS ORDINARY ROUTE
- 3 OPTIONAL AI SERVICE
- 4 HUMAN HANDOFF + APPEAL
- 5 SHUTDOWN + RECOVERY + REVIEW

THREE NON-NEGOTIABLES

- Accessibility and ordinary routes never depend on AI
- No default face, identity or continuous tracking
- Operators and on-site staff retain stop authority

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01

Ordinary route always exists

02

Visible and reachable human service

03

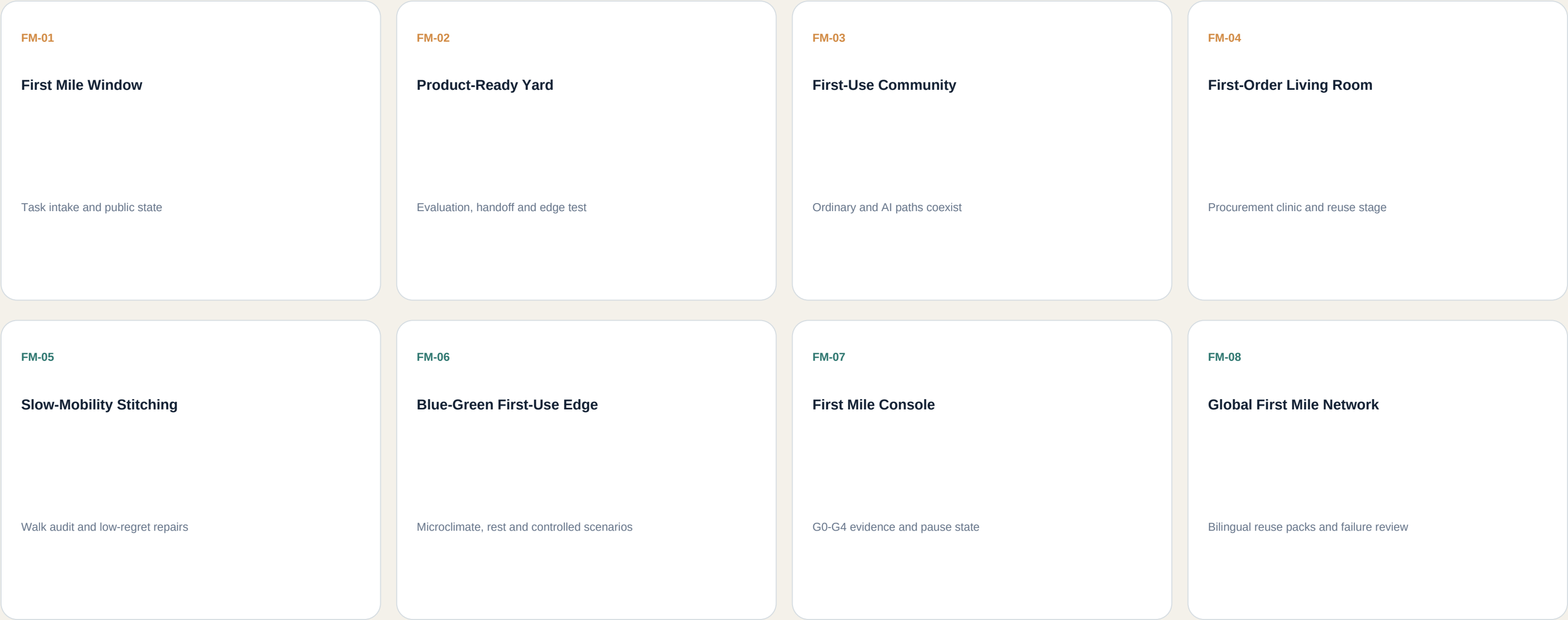
Data minimization

04

Low-speed tests are bounded in place and time

Eight Low-Regret Projects: Operations Before Construction

Small, reversible and staged; no device procurement without an owner, staff roster, maintenance and exit resource.

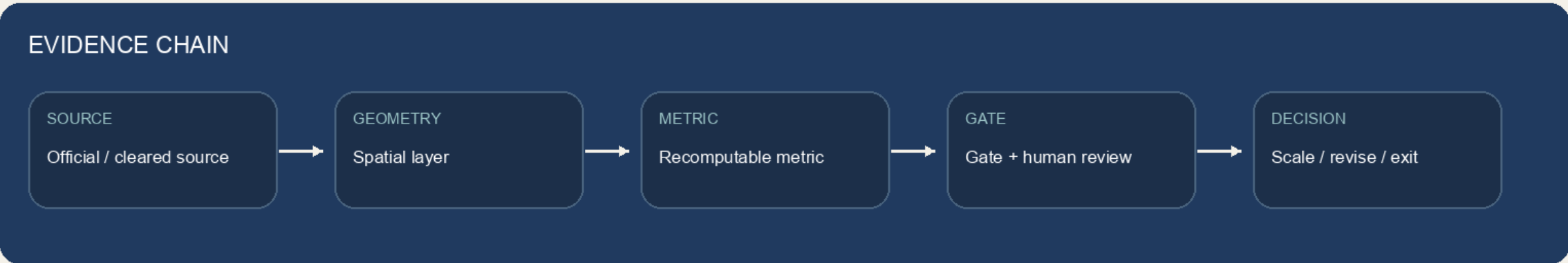


Known Spatial Metrics Recompute; Adoption Outcomes Stay Unknown

Provisional geometry supports concept comparison, not an official redline or statutory target.

METRICS + EVIDENCE: PROVE ADOPTION BEFORE SCALE

SPATIAL METRICS USE PROVISIONAL GEOMETRY; OPERATING OUTCOMES REMAIN UNKNOWN BEFORE A REAL PILOT



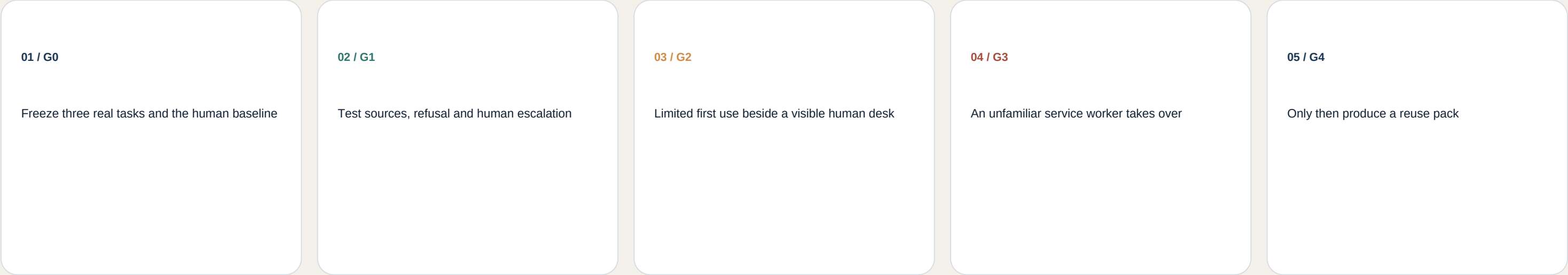
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The First 90 Days: Verify One First Real User

Start with the AI Startup Service Copilot at AI Origin; do not fill the belt, buy robots or call traffic adoption.



From Claim to Decision: Leave Reviewable Evidence

Sources, GeoJSON, metrics, matrices and self-checks form the authority layer; failures and pauses remain public evidence.

